

```
a = 0
b = 2
c = 10
for number in range(3, 15, 3):
    a *= number
    if number == 12:
        b += 5
    c = c * number
```

- a. **What is the value of b after this for loop is run? Why?**
 - i. b = 7 because 12 is included in the range and adds 5 to 2
- b. **If I change the number in the conditional if statement from 12 to 15 will that change the value of b? Yes or no? Explain.**
 - i. Yes, because 15 is not included in the range because range ends before the stop value, so 5 will not be added to b, meaning b = 2
- c. **Is variableName = variableName + 1 in Python equivalent to variableName += 1? Yes or no? Explain. If yes, why aren't a and c the same value? Explain.**
 - i. Yes, because they are both updating the value of variableName to variableName + 1.
 - ii. a and c have different initial values